Introduction to Networking and the OSI ModelTheory

1.Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

ANS:

Wireshark Packet Capture Report

Aim: To capture network packets using Wireshark and identify the protocols used when visiting a website.

Steps Performed:

1. Download and install Wireshark from the official website: https://www.wireshark.org/
2. Open Wireshark and select the main network interface, such as Wi-Fi or Ethernet.
3. Click Start Capturing Packets.
4. Open a browser and visit instagram.com.
5. Return to Wireshark and click Stop.
6. Observe the protocol names in the Protocol column.

Protocols Observed (Example):

No.	Protocol	Purpose
1	DNS	Resolves the website's domain name to an IP address.
2	TCP	Provides reliable data transmission.
3	TLS	Encrypts communication with HTTPS websites.
4	QUIC	Supports fast, secure communication, often over UDP.
5	UDP	Transfers datagrams and supports protocols such as QUIC.
6	ARP	Helps find the MAC address associated with an IPv4 address on the local network.

**2.Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.).

Hint: Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.**

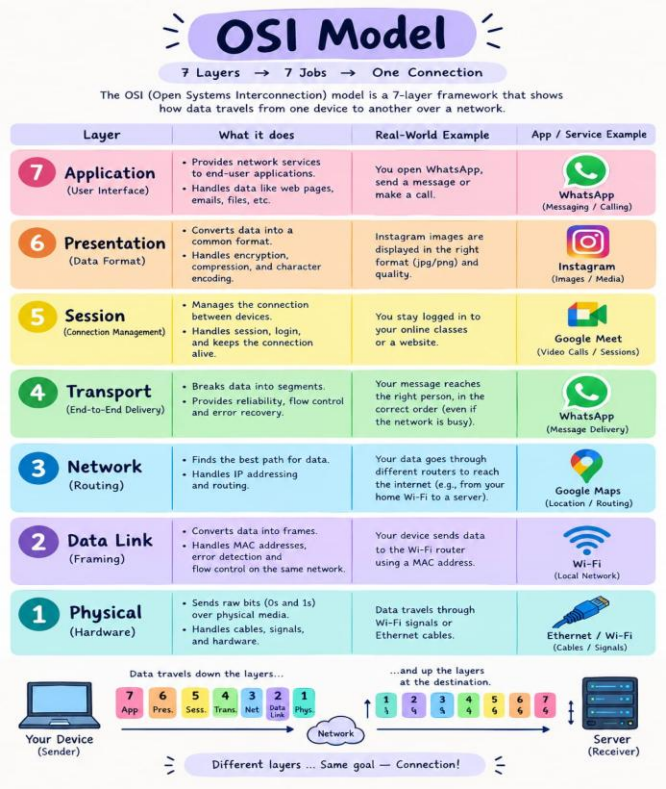
ANS:

Wireshark Packet Analysis – OSI Layers

No.	Protocol	Packet Details	OSI Layer
1	DNS	Resolves the website name to an IP address.	Layer 7 – Application
2	TCP	Provides reliable data transmission between devices.	Layer 4 – Transport
3	Ethernet	Transfers frames using MAC addresses on the local network.	Layer 2 – Data Link

3.Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

ANS:



4.Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own

**words.

Hint: Think about how your message is typed, packaged, sent over the internet, and received.**

ANS:

How a WhatsApp Message Travels Through the OSI Model

When we send a message on WhatsApp, it passes through different OSI layers to reach the receiver.

OSI Layer	**What Happens**
7. Application	We type a message and press Send in WhatsApp.
6. Presentation	The message data is formatted and encrypted to protect its contents.
4. Transport	Transport protocols help deliver data between devices. TCP or UDP may be used, depending on the communication method.
3. Network	IP addresses help route the data across networks toward its destination.
2. Data Link	Data is placed into frames for transfer across the local network, such as Wi-Fi.
1. Physical	The data travels as radio signals over Wi-Fi or as electrical or optical signals through cables.